

BLINK WITHOUT A CHIP!

The Magic Electricity Storage Adventure ✨

Hello, Future Electronics Wizard! 🙌

Have you ever seen fairy lights blink on and off? Usually, they need a tiny computer chip "brain" to tell them what to do. But today, we are going to perform a magic trick. We will build a circuit that blinks completely on its own—with **no computer chips allowed!** Let's meet our magical team.



Meet Your Circuit Team

In your toolkit box, you have 5 special friendly characters. Each one has a secret job to do:

COMPONENT	ADVENTURE IDENTITY	THEIR SECRET LAB JOB
9V Battery	The Water Pump	Pushes electricity out into the wires to get everything moving!
Resistor	The Squeeze Tap	Slows down the electricity so it can only travel drop... by... drop!
Capacitor	The Magic Bucket	Catches the electricity drops and stores them up safely until full.
Transistor	The Gate Guard	A smart switch that stands at the gate and snaps open when pushed!
LED Light	The Shiny Lantern	Glow super bright to show us that our circuit magic is working!



LED Secret: Check Your Legs!

Look very closely at your LED light bulb. It has two metal wire legs that must go the right way:

- **The Long Leg (+):** This is the happy, positive leg. It always wants to face toward the positive side of your battery.

- **The Short Leg (-):** This is the negative leg. It points toward the negative side of your battery.

⚠ **Wizard Tip:** If you plug the LED in backwards, it won't break, but the light will stay completely dark!

The 3 Steps of the Blinking Loop

When we connect our characters together, they create a perfect, repeating cycle:

Step 1: Squeezing and Filling 💧

The **Battery (Pump)** pushes electricity out. It hits the **Resistor (Squeeze Tap)**, which slows it down. The electricity slowly flows into the **Capacitor (Magic Bucket)** drop... by... drop. Right now, the LED lantern is completely **OFF**.

Step 2: The Big Splash! 🪣

Suddenly, the **Capacitor (Magic Bucket)** gets filled right to the top! It can't hold any more drops, so it splashes its extra stored power directly against the **Transistor (Gate Guard)**.

Step 3: Flash and Reset! ⚡

Boom! The **Gate Guard** feels the splash and instantly throws the gate wide open. All the electricity rushes into the **LED Light**, making it flash **ON** with a bright glow! But now the bucket is empty, the gate slams shut, the LED goes **OFF**, and Step 1 starts all over again!

Lab Safety Rules for Young Scientists

- **No Wire Tangling:** Never let the red battery wire and black battery wire touch each other directly. It makes the battery super hot!
- **Keep Hands Dry:** Always wash and dry your hands before building. Water on your fingers can block the electric path.

Creative Brain Challenge!

What do you think would happen if we swapped our Resistor for a **much wider tap** that lets electricity run through super fast?

Would our magic bucket fill up quicker or slower? Would the LED light blink faster or slower? Think about your theory, test it on your lab bench, and share your discovery!